

AD 2 AERODROMES

RJCM AD 2.1 AERODROME LOCATION INDICATOR AND NAME

RJCM - MEMANBETSU

RJCM AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	ARP coordinates and site at AD	435250N / 1440951E 175° / 1.25km from RWY 18 THR
2	Direction and distance from (city)	9.7nm SSW ABASHIRI
3	Elevation/ Reference temperature	109FT / 26°C (2004-2008)
4	Geoid undulation at AD ELEV PSN	99FT
5	MAG VAR/ Annual change	9° W(2009) / 2.1'E
6	AD Administration, address, telephone, telefax, telex, AFS, e-mail and/or Web-site addresses	Hokkaido Airports Co.,Ltd. Memanbetsu Airport Office 256-3, Chuo, Memanbetsu, Ozora-cho Abashiri-gun, Hokkaido TEL: 0152-74-2222 FAX: 0152-74-3674 e-MAIL: hap-mmb-unjyo@hokkaido-airports.co.jp
7	Types of traffic permitted(IFR/VFR)	IFR/VFR
8	Remarks	Memanbetsu Airport Branch (CAB) 256, Chuo, Memanbetsu, Ozora-cho Abashiri-gun, Hokkaido TEL:0152-74-2673

RJCM AD 2.3 OPERATIONAL HOURS

1	AD Administration	2300-1200
2	Customs and immigration	On request Customs: 0154-22-3730 Immigration: 0154-22-2430
3	Health and sanitation	Quarantine(human): On request(0154-23-3340) Quarantine(animal, plant): Nil
4	AIS Briefing Office	Nil
5	ATS Reporting Office(ARO)	Nil
6	MET Briefing Office	H24 (NEW CHITOSE)
7	ATS	2300-1200
8	Fuelling	2300-1200
9	Handling	2300-1200
10	Security	2300-1200
11	De-icing	Nil
12	Remarks	Nil

RJCM AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	All the modern institutions that deal with the weight thing to a A306 type freighter
2	Fuel/ oil types	JET A-1
3	Fuelling facilities/ capacity	Fuel truck refueling, 22L/sec
4	De-icing facilities	Nil
5	Hangar space for visiting aircraft	Nil
6	Repair facilities for visiting aircraft	Nil
7	Remarks	Nil

RJCM AD 2.5 PASSENGER FACILITIES

1	Hotels	Nil
2	Restaurants	At airport
3	Transportation	Buses and taxis
4	Medical facilities	Hospital in Ozora-town, 5km from AP
5	Bank and Post Office	Nil
6	Tourist Office	At airport
7	Remarks	Nil

RJCM AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	AD category for fire fighting	CAT 8
2	Rescue equipment	Chemical fire fighting truck x 3 Emergency medical equipments conveyance truck x 1
3	Capability for removal of disabled aircraft	INFO on EQPT AVBL in AP: Ask AD administration MAX type of ACFT removed by EQPT: B767-300ER CTC details: Hokkaido Airports Co.,Ltd. Memanbetsu Airport Office TEL: 0152-74-2222 Mail: hap-mmb-unjyo@hokkaido-airports.co.jp
4	Remarks	Nil

RJCM AD 2.7 SEASONAL AVAILABILITY-CLEARING

1	Types of clearing equipment	Snow removal equipment: 42
2	Clearance priorities	1) RWY 18/36, T1, T6, P1-P6, Apron A 2) T2-T5, TB, Apron B
3	Remarks	Nil

RJCM AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS DATA

1	Apron surface and strength	Apron A : Surface:Cement-concrete, Strength:PCR 1132/R/B/W/T Apron B : Surface:Cement-concrete, Strength:PCR 257/R/B/W/T
2	Taxiway width, surface and strength	T1-T6, P1-P6 : Surface:Asphalt-concrete, Width:30m, Strength:PCR 1170/F/C/X/T TB : Surface:Asphalt-concrete, Width:9m, Strength:PCR 273/F/C/X/T
3	ACL and elevation	Not available
4	VOR checkpoints	Not available
5	INS checkpoints	(Spot NR) 1 : 435257.24N/1440933.47E 2 : 435255.56N/1440933.89E 3 : 435253.47N/1440934.16E 4 : 435251.21N/1440934.53E 5 : 435248.95N/1440934.82E 6 : 435246.85N/1440935.01E
6	Remarks	Nil

RJCM AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand ID signs, TWY guide lines and Visual docking/ parking guidance system of aircraft stands	Aircraft stand ID sign : Nil ACFT stand taxi lane marking : See AD2.24. AD chart Visual docking guidance system : Nil
2	RWY and TWY markings and LGT	RWY: RWY18/36 (Marking) RWY designation, RWY CL, RWY THR, Aiming point, TDZ, RWY side stripe (LGT) RCLL, REDL, RTHL, RENL, RTZL, RWY DIST marker LGT, WBAR TWY: ALL (Marking) TWY CL, TWY side stripe (LGT) TWY edge LGT TWY: T1-T6 (Marking) RWY HLDG PSN, Mandatory instruction (LGT) TWY CL LGT, RWY guard LGT, Taxiing guidance sign TWY: P1-P6 (LGT) TWY CL LGT TWY: TB (Marking) RWY HLDG PSN, Mandatory instruction (LGT) RWY guard LGT, Taxiing guidance sign
3	Stop bars	Nil
4	Remarks	(Marking) Overrun area (LGT) Apron flood LGT

RJCM AD 2.10 AERODROME OBSTACLES

In Area2 See Obstacle data

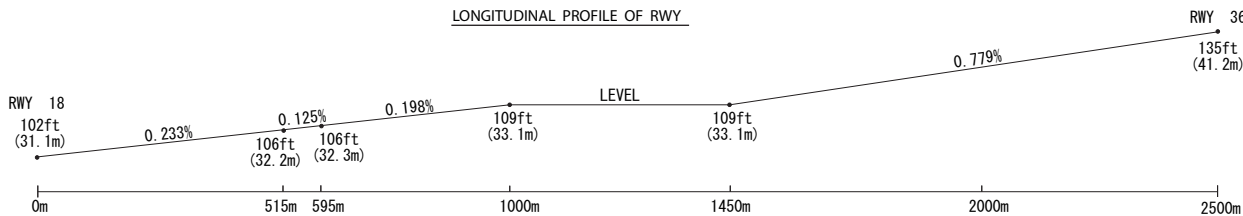
In Area3 To be developed

RJCM AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Associated MET Office	NEW CHITOSE
2	Hours of service MET Office outside hours	H24 (NEW CHITOSE)
3	Office responsible for TAF preparation Periods of validity	NEW CHITOSE 30 Hours
4	Trend forecast Interval of issuance	Nil
5	Briefing/ consultation provided	Briefing is available upon inquiry at NEW CHITOSE
6	Flight documentation Language(s) used	C En
7	Charts and other information available for briefing or consultation	S ₆ , U ₈₅ , U ₇ , U ₅ , U ₃ , U ₂₅ , U ₂ /T _r , P _s , P ₅ , P ₃ , P ₂₅ , P _{SWE} , P _{SWF} , P _{SWG} , P _{SWI} , P _{SWM} , P _{SW} (domestic), E, C, W _E , W _F , W _G , W _I , W, N
8	Supplementary equipment available for providing information	Nil
9	ATS units provided with information	TWR
10	Additional information(limitation of service, etc.)	Nil

RJCM AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations RWY NR	TRUE BRG	Dimensions of RWY(M)	Strength(PCR) and Surface of RWY	THR coordinates THR geoid undulation	THR elevation and highest elevation of TDZ of precision APP RWY
1	2	3	4	5	6
18	174.66°	2500x45	PCR 1027/F/D/X/T Asphalt-concrete	435330.51N/1440945.38E 98.8FT	THR ELEV: 102.0FT TDZ ELEV: 108FT
36	354.66°	2500x45		435209.85N/1440955.80E 98.8FT	THR ELEV: 135.2FT TDZ ELEV: 131FT
Slope of RWY		Strip Dimensions(M)	RESA (Overrun) Dimensions (M)		Remarks
7		10	11		14
See below figure		2620x300	190×(MNM:140 MAX:300)*		RWY Grooving 2500m x 45m
		2620x300	90×(MNM:90 MAX:300)* *For detail, ask airport administrator		



RJCM AD 2.13 DECLARED DISTANCES

RWY Designator	TORA (m)	TODA (m)	ASDA (m)	LDA (m)	Remarks
1	2	3	4	5	6
18	2500	2500	2500	2500	Nil
36	2500	2500	2500	2500	Nil

RJCM AD 2.14 APPROACH AND RUNWAY LIGHTING

RWY Designator	APCH LGT type LEN INTST	RTHL Color WBAR	PAPI (VASIS) Angle DIST FM THR MEHT	RTZL LEN	RCLL LEN Spacing Color INTST	REDL LEN Spacing Color INTST	RENL Color WBAR	STWL LEN Color
1	2	3	4	5	6	7	8	9
18	PALS (CAT I) 900m LIH	Green Green	PAPI 3.0°/Left 404.5m 65.6ft	900m	2500m 30m Coded color (White/Red) LIH	2500m 60m Coded color (White/Yellow) LIH	Red	Nil(*1)
36	PALS (CAT I) 900m LIH	Green Green	PAPI 3.0°/Left 499.2m 65.6ft	900m	2500m 30m Coded color (White/Red) LIH	2500m 60m Coded color (White/Yellow) LIH	Red	Nil(*1)
Remarks								
10								
Overrun area edge LGT(LEN:60m, Color:Red)(*1)								

RJCM AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	ABN/IBN location, characteristics and hours of operation	ABN: 435259N /1440926E , ALTN FLG(2)WG EV 4.3SEC, HO
2	LDI location and LGT Anemometer location and LGT	LDI : Nil Anemometer : RWY 18: 285m from RWY 18 THR RWY 36: 289m from RWY 36 THR
3	TWY edge and center line lighting	TWY edge and center line lights installed, see AD2.9
4	Secondary power supply / switch-over time	Within 1sec : REDL, RENL, RTHL, WBAR, RCLL, Overrun area edge LGT Within 15sec : Other LGT
5	Remarks	WDI LGT

RJCM AD 2.16 HELICOPTER LANDING AREA

Nil

RJCM AD 2.17 ATS AIRSPACE

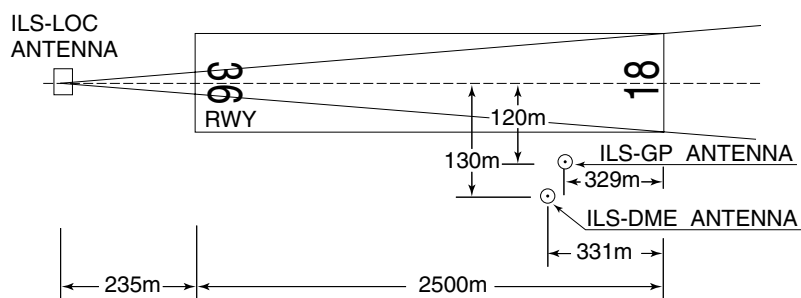
Designation and lateral limits		Vertical limits (ft)	Airspace classification	ATS unit call sign Language	Remarks
1		2	3	4	6
Memambetsu CTR	Area within a radius of 5nm(9km) of Memambetsu ARP (4353N/14410E)	3000 or below	D	Memambetsu TWR En	
Hidaka ACA	See RJEC attached chart		E	Hidaka APP En	

RJCM AD 2.18 ATS COMMUNICATION FACILITIES

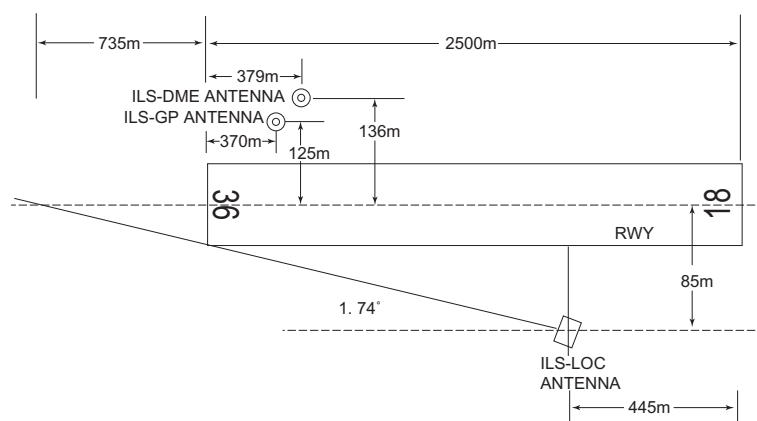
Service designation	Call sign	Frequency	Hours of operation	Remarks
1	2	3	4	5
APP	Hidaka Approach	128.325MHz 246.1MHz 134.55MHz 121.5MHz (E) 243.0MHz (E)	2230 - 1200	
TWR	Memambetsu tower	118.85MHz(1) 126.2MHz	2300 - 1200 (1)Primary	

RJCM AD 2.19 RADIO NAVIGATION AND LANDING AIDS

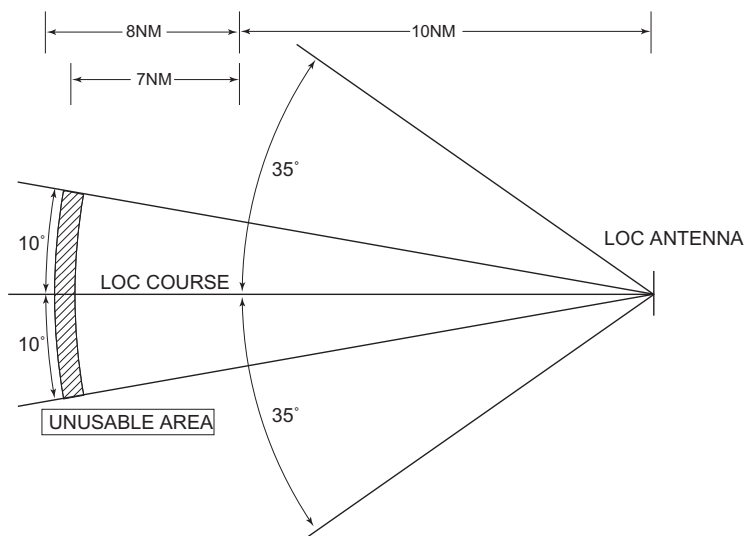
Type of aid (VOR declination)	ID	Frequency	Hours of operation	Position of transmitting antenna coordinates	Elevation of DME transmitting antenna	Remarks
1	2	3	4	5	6	7
VOR (10°W/2024)	TBE	110.85MHz	H24	435305.67N/ 1440958.26E		
DME	TBE	1132MHz (CH-45Y)	H24	435305.67N/ 1440958.26E	131ft	DME Unusable: 150° - 160° beyond 30nm BLW 7000ft. 300° - 320° beyond 35nm BLW 5000ft. 330° - 340° beyond 35nm BLW 3000ft.
ILS-LOC 18	ITB	110.1MHz	2300 - 1200	435202.26N/ 1440956.74E		LOC: 235m (771ft) away FM RWY 36 THR, BRG (MAG) 184°
ILS-GP 18	-	334.4MHz	2300 - 1200	435320.20N/ 1440952.07E		GP: 329m (1079ft) inside FM RWY 18 THR, 120m (394ft) E of RCL. Angle 3.0°, HGT of ILS reference datum 16.5m (54ft)
ILS-DME 18	ITB	999MHz (CH-38X)	2300 - 1200	435320.16N/ 1440952.39E	119ft	DME: 331m (1086ft) inside FM RWY 18 THR, 130m(427ft) E of RCL.
ILS-LOC 36	IHM	110.3MHz	2300 - 1200	435316.40N/ 1440950.98E		LOC: 445m(1460ft) inside FM RWY 18 THR, 85m(279ft) E of RCL. LOC offset angle 1.74° BRG(MAG) 5.90° LOC unusable: beyond 17nm from LOC antenna
ILS-GP 36	-	335MHz	2300 - 1200	435221.40N/ 1440948.65E		GP: 370m(1214ft) inside FM RWY 36 THR, 125m(410ft) W of RCL. HGT of ILS REF datum: 16.5m(54ft). GP angle 3.0°.
ILS-DME 36	IHM	1001MHz (CH-40X)	2300 - 1200	435221.65N/ 1440948.07E	137ft	DME: 379m(1243ft) inside FM RWY 36, 136m(446ft) W of RCL.
MSAS		1575.42MHz	H24			Transmitting antennas are satellite based.

ILS for RWY18

REMARKS : 1. LOC beam BRG(MAG) 184°
 2. HGT of ILS REF datum 16.5m(54ft)
 3. GP Angle 3.0°
 4. ELEV of ILS-DME 36.2m(119ft)

ILS for RWY36

REMARKS : 1. LOC OFFSET ANGLE 1.74°
 2. LOC beam BRG(MAG) 5.90°
 3. HGT of ILS REF datum 16.5m(54ft)
 4. GP Angle 3.0°
 5. ELEV of ILS-DME 41.7m (137ft)



LOC unusable in the following area : BEY 17NM FM LOC ANT.

RJCM AD 2.20 LOCAL TRAFFIC REGULATIONS

1. Airport regulations

PPR
Prior permission is required for transient aircraft due to parking congestion except scheduled and/or emergency flight.
Tel: Hokkaido Airports Co.,Ltd. Memanbetsu Airport Office 0152-74-2222

2. Taxiing to and from stands

Nil

3. Parking area for small aircraft(General aviation)

Nil

4. Parking area for helicopters

Nil

5. Apron - taxiing during winter conditions

Nil

6. Taxiing - limitations

Nil

7. School and training flights - technical test flights - use of runways

Nil

8. Helicopter traffic - limitation

Nil

9. Removal of disabled aircraft from runways

Terms of use:
As a rule, when submitting notification of airport use prior to flight operations, a "Consent Form for Airport Use" and an "aircraft operator's disabled aircraft removal operation plan" must be prepared and submitted to airport administrator. In addition, when preparing these documents in advance, coordinating with point of contact in 2.6.3 must be confirmed.

RJCM AD 2.21 NOISE ABATEMENT PROCEDURES

Nil

RJCM AD 2.22 FLIGHT PROCEDURES**1.TAKE OFF MINIMA**

	RWY	ACFT CAT	REDL & RCLL		REDL or RCLL or RCL Marking		NIL (DAYTIME ONLY)	
			RVR	VIS	RVR	VIS	RVR	VIS
Multi-Engine ACFT with TKOF ALTN AP FILED	18	A,B,C,D	400m	400m	400m	400m	-	500m
	36	A,B,C,D	400m	400m	400m	400m	-	500m
OTHER	18	A,B,C,D	AVBL LDG MINIMA					
	36	A,B,C,D						

2. Lost communication procedures for arrival aircraft under radar navigational guidance

If radio communications with Hidaka Approach are lost for 1 minute, squawk Mode A/3 Code 7600 and;

- (I) 1. Contact Memanbetsu Tower.
 2. If unable, proceed in accordance with visual flight rules.
 3. If unable, proceed to MEMANBETSU VOR/DME at last assigned altitude or 4,000 feet whichever is higher, and execute instrument approach.
- (II) Procedures other than above will be issued when situation requires.

RJCM AD 2.23 ADDITIONAL INFORMATION

双方向に設置されている ILS の輻射について

ILS 18/36

RWY18 および 36 ILS は同時に輻射する。

Two separate ILS radiate at opposite ends of a single runway

ILS 18/36

RWY18 and 36 ILS radiate simultaneously.

RJCM AD 2.24 CHARTS RELATED TO AN AERODROME

Aerodrome/Heliport Chart
 Standard Departure Chart (MEMANBETSU REVERSAL)
 Standard Departure Chart (MENIB-RNAV)
 Standard Departure Chart (NICOL-RNAV)
 Standard Departure Chart (NULKI-RNAV)
 Instrument Approach Chart (ILS Z or LOC Z RWY18)
 Instrument Approach Chart (ILS Y or LOC Y RWY18)
 Instrument Approach Chart (ILS X RWY18)
 Instrument Approach Chart (ILS or LOC RWY36)
 Instrument Approach Chart (VOR RWY18)
 Instrument Approach Chart (VOR RWY36)
 Instrument Approach Chart (RNP Z RWY18)
 Instrument Approach Chart (RNP Y RWY18 (AR))
 Instrument Approach Chart (RNP RWY36)
 Other Chart (Visual REP)
 Other Chart (LDG CHART)
 Other Chart (MVA CHART)

[illegible]

STANDARD DEPARTURE CHART-INSTRUMENT

RJCM / MEMANBETSU

SID

MEMANBETSU REVERSAL FOUR DEPARTURE

RWY 18 : Climb via TBE R188 to TBE 5.0DME, turn left,...

RWY 36 : Climb via TBE R001 to TBE 5.0DME, turn right,...

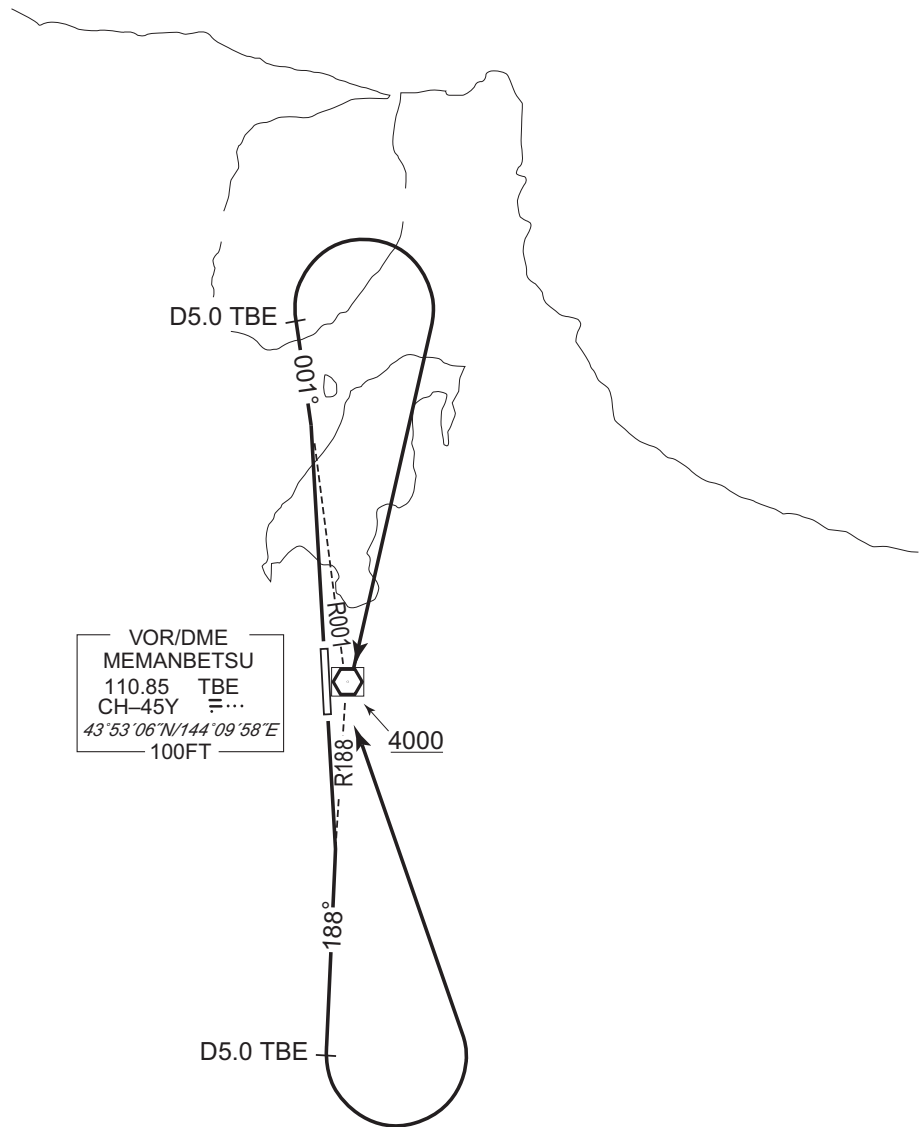
....direct to TBE VOR/DME.

Cross TBE VOR/DME at or above 4000FT.

Note RWY 18 : 5.0% climb gradient required up to 300FT.

OBST ALT 176FT located at 0.2NM 131° FM end of RWY18.

OBST ALT 216FT located at 0.4NM 148° FM end of RWY18.

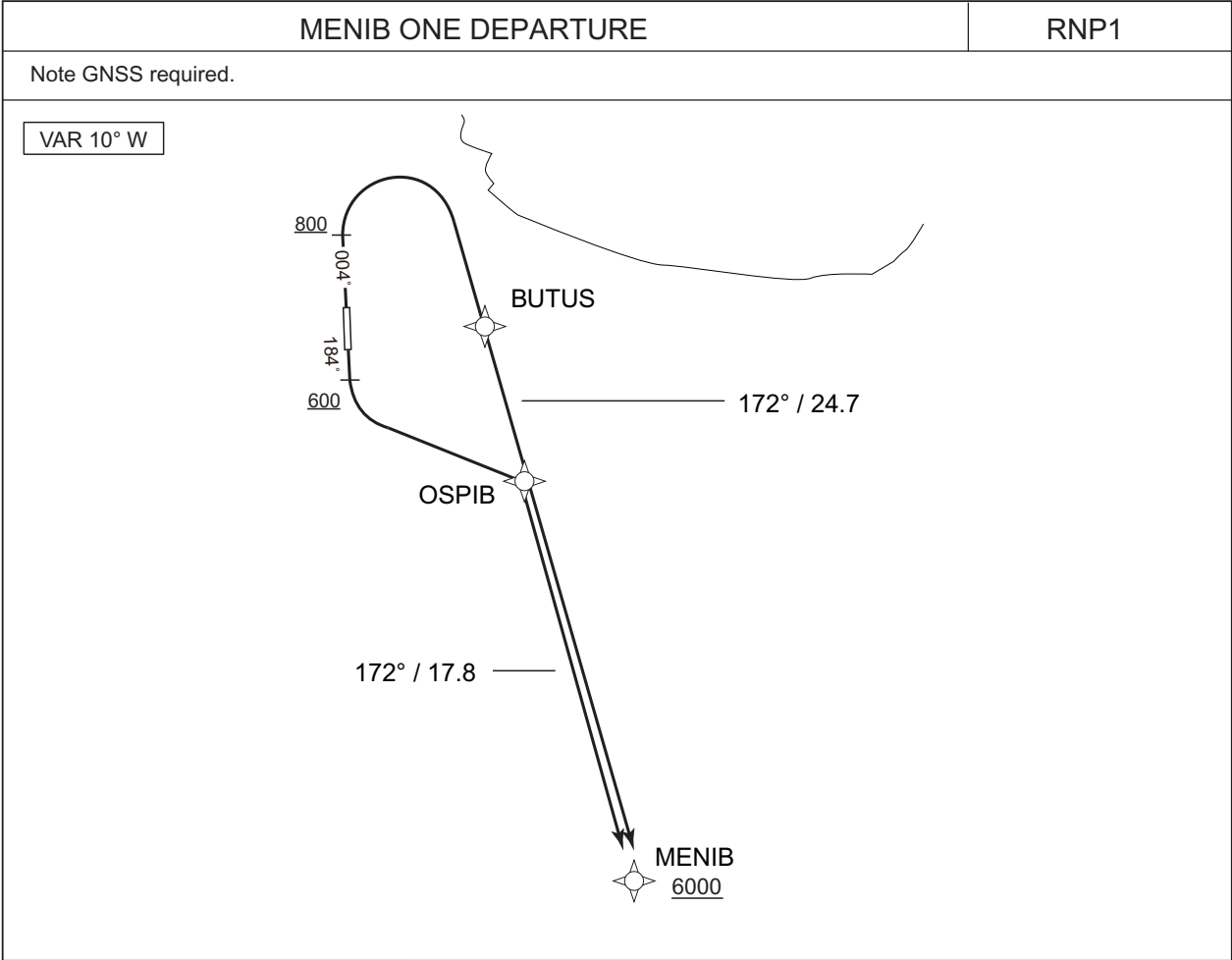


CHANGE : PROC renamed. PROC course. Note added.

STANDARD DEPARTURE CHART-INSTRUMENT

RJCM / MEMANBETSU

RNAV SID



RWY18 : Climb on HDG184° at or above 600FT, turn left direct to OSPIB, to MENIB at or above 6000FT.
RWY36 : Climb on HDG004° at or above 800FT, turn right direct to BUTUS, to MENIB at or above 6000FT.
Note RWY18 : 5.0% climb gradient required up to 2300FT.
OBST ALT 3314FT located at 12.2NM 153° FM end of RWY18.

RWY18

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	—	—	184 (174.7)	-9.6	—	—	+600	—	—	RNP1
002	DF	OSPIB	—	—	-9.6	—	L	—	—	—	RNP1
003	TF	MENIB	—	172 (162.3)	-9.6	17.8	—	+6000	—	—	RNP1

RWY36

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	—	—	004 (354.7)	-9.6	—	—	+800	—	—	RNP1
002	DF	BUTUS	—	—	-9.6	—	R	—	—	—	RNP1
003	TF	MENIB	—	172 (162.3)	-9.6	24.7	—	+6000	—	—	RNP1

CHANGE : Description of latitude and longitude.

STANDARD DEPARTURE CHART-INSTRUMENT

RJCM / MEMANBETSU

RNAV SID

Waypoint Coordinates

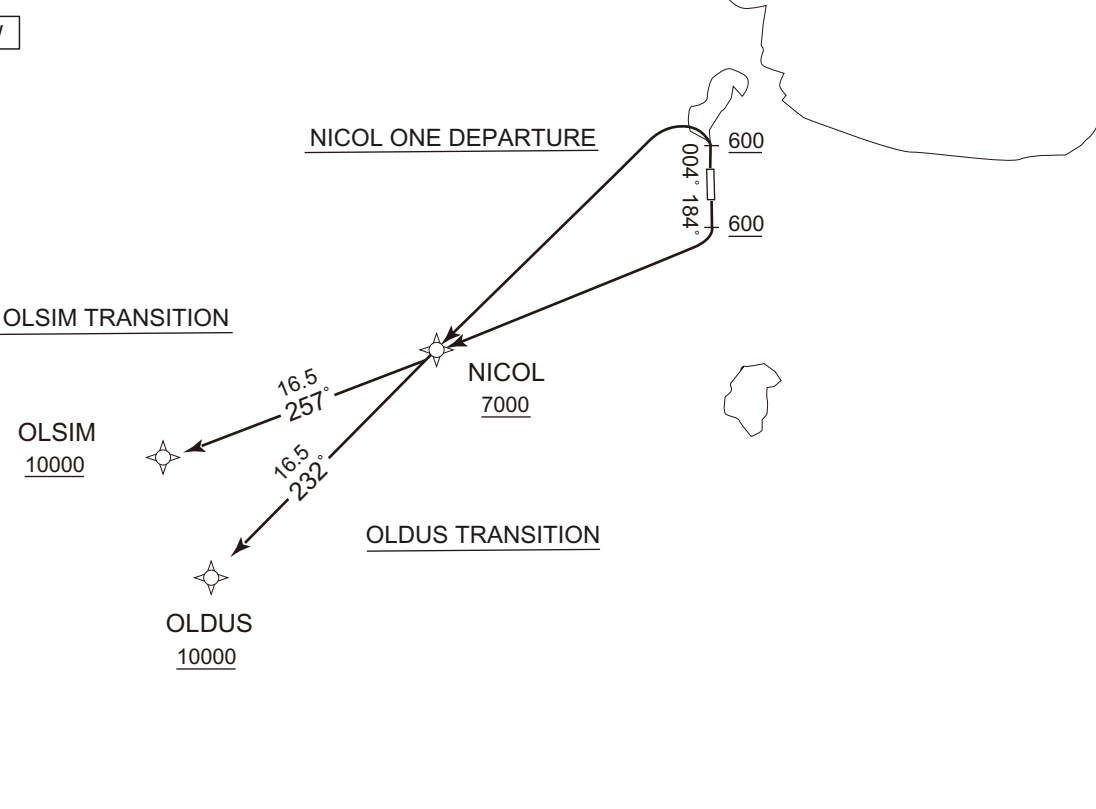
Waypoint Identifier	Coordinates
OSPIB	434628.5N / 1442055.9E
BUTUS	435301.9N / 1441802.1E
MENIB	432929.4N / 1442822.9E

CHANGE : Waypoint Coordinates added.

STANDARD DEPARTURE CHART-INSTRUMENT

RJCM / MEMANBETSU

RNAV SID and TRANSITION

NICOL ONE DEPARTURE OLSIM TRANSITION / OLDUS TRANSITION		RNP1
Note GNSS required.		
VAR 10° W 		
NICOL ONE DEPARTURE RWY18 : Climb on HDG184° at or above 600FT, turn right direct to NICOL at or above 7000FT. RWY36 : Climb on HDG004° at or above 600FT, turn left direct to NICOL at or above 7000FT. Note RWY36 : 5.0% climb gradient required up to 600FT. OBST ALT 1181FT located at 6.3NM 312° FM end of RWY36.		
OLSIM TRANSITION From NICOL at or above 7000FT, to OLSIM at or above 10000FT.		
OLDUS TRANSITION From NICOL at or above 7000FT, to OLDUS at or above 10000FT.		

CHANGE : Description of latitude and longitude.

STANDARD DEPARTURE CHART-INSTRUMENT

RJCM / MEMANBETSU

RNAV SID and TRANSITION

NICOL ONE DEPARTURE

RWY18

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	184 (174.7)	-9.6	-	-	+600	-	-	RNP1
002	DF	NICOL	-	-	-9.6	-	R	+7000	-	-	RNP1

RWY36

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	-	-	004 (354.7)	-9.6	-	-	+600	-	-	RNP1
002	DF	NICOL	-	-	-9.6	-	L	+7000	-	-	RNP1

OLSIM TRANSITION

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	NICOL	-	-	-9.6	-	-	+7000	-	-	RNP1
002	TF	OLSIM	-	257 (247.6)	-9.6	16.5	-	+10000	-	-	RNP1

OLDUS TRANSITION

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	NICOL	-	-	-9.6	-	-	+7000	-	-	RNP1
002	TF	OLDUS	-	232 (222.9)	-9.6	16.5	-	+10000	-	-	RNP1

CHANGE : Waypoint Coordinates added.

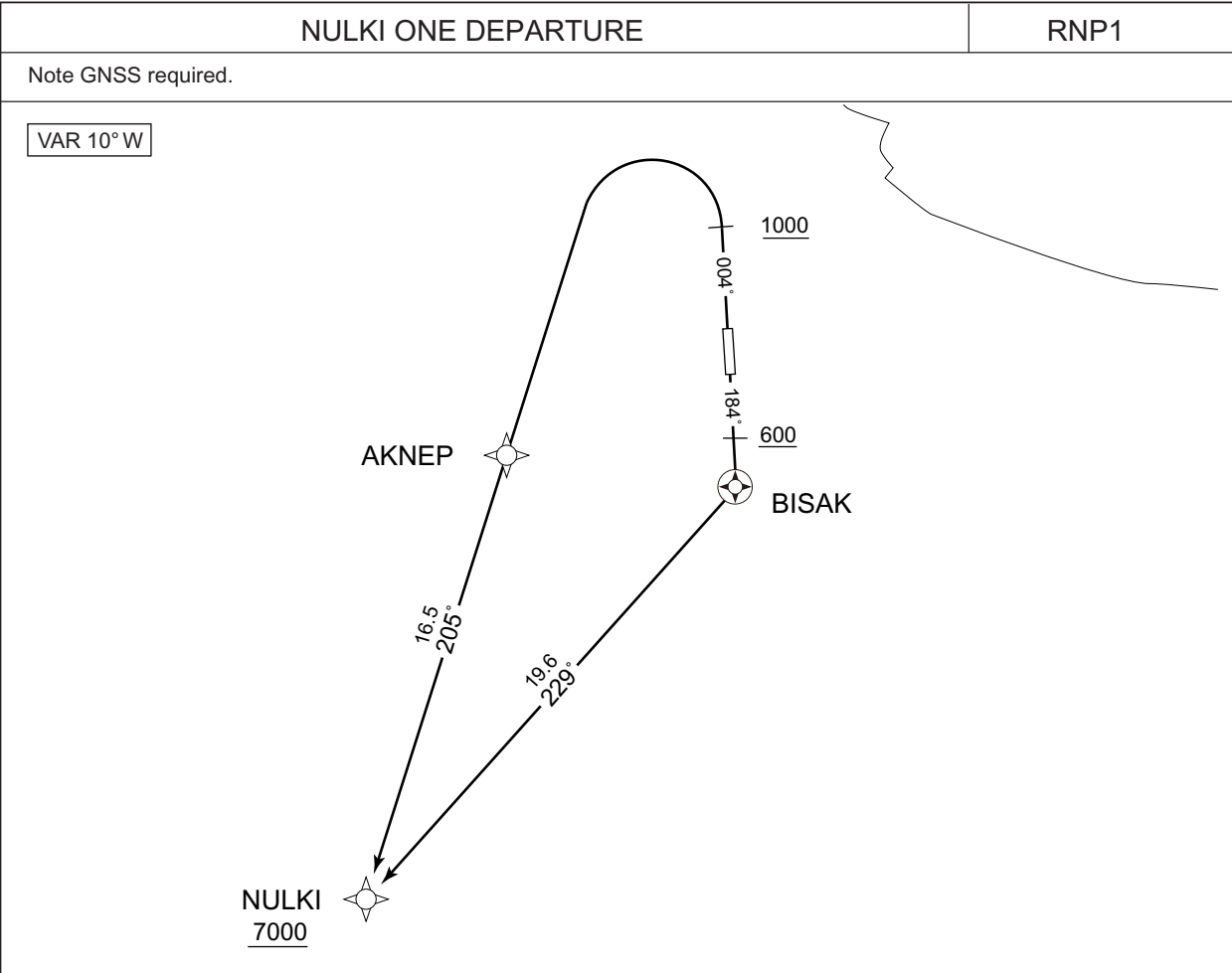
Waypoint Coordinates

Waypoint Identifier	Coordinates
NICOL	433912.0N / 1434158.7E
OLSIM	433251.7N / 1432055.3E
OLDUS	432703.7N / 1432629.1E

STANDARD DEPARTURE CHART-INSTRUMENT

RJCM / MEMANBETSU

RNAV SID



RWY18 : Climb on HDG184° at or above 600FT, direct to BISAK, to NULKI at or above 7000FT.
RWY36 : Climb on HDG004° at or above 1000FT, turn left direct to AKNEP, to NULKI at or above 7000FT.
Note RWY36 : 3.9% climb gradient required up to 1500FT.
OBST ALT 1673FT located at 8.2NM 320° FM end of RWY36.

RWY18

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	—	—	184 (174.7)	-9.6	—	—	+600	—	—	RNP1
002	DF	BISAK	Y	—	-9.6	—	—	—	—	—	RNP1
003	TF	NULKI	—	229 (219.6)	-9.6	19.6	—	+7000	—	—	RNP1

RWY36

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	VA	—	—	004 (354.7)	-9.6	—	—	+1000	—	—	RNP1
002	DF	AKNEP	—	—	-9.6	—	L	—	—	—	RNP1
003	TF	NULKI	—	205 (194.9)	-9.6	16.5	—	+7000	—	—	RNP1

CHANGE : Description of latitude and longitude.

STANDARD DEPARTURE CHART-INSTRUMENT

RJCM / MEMANBETSU

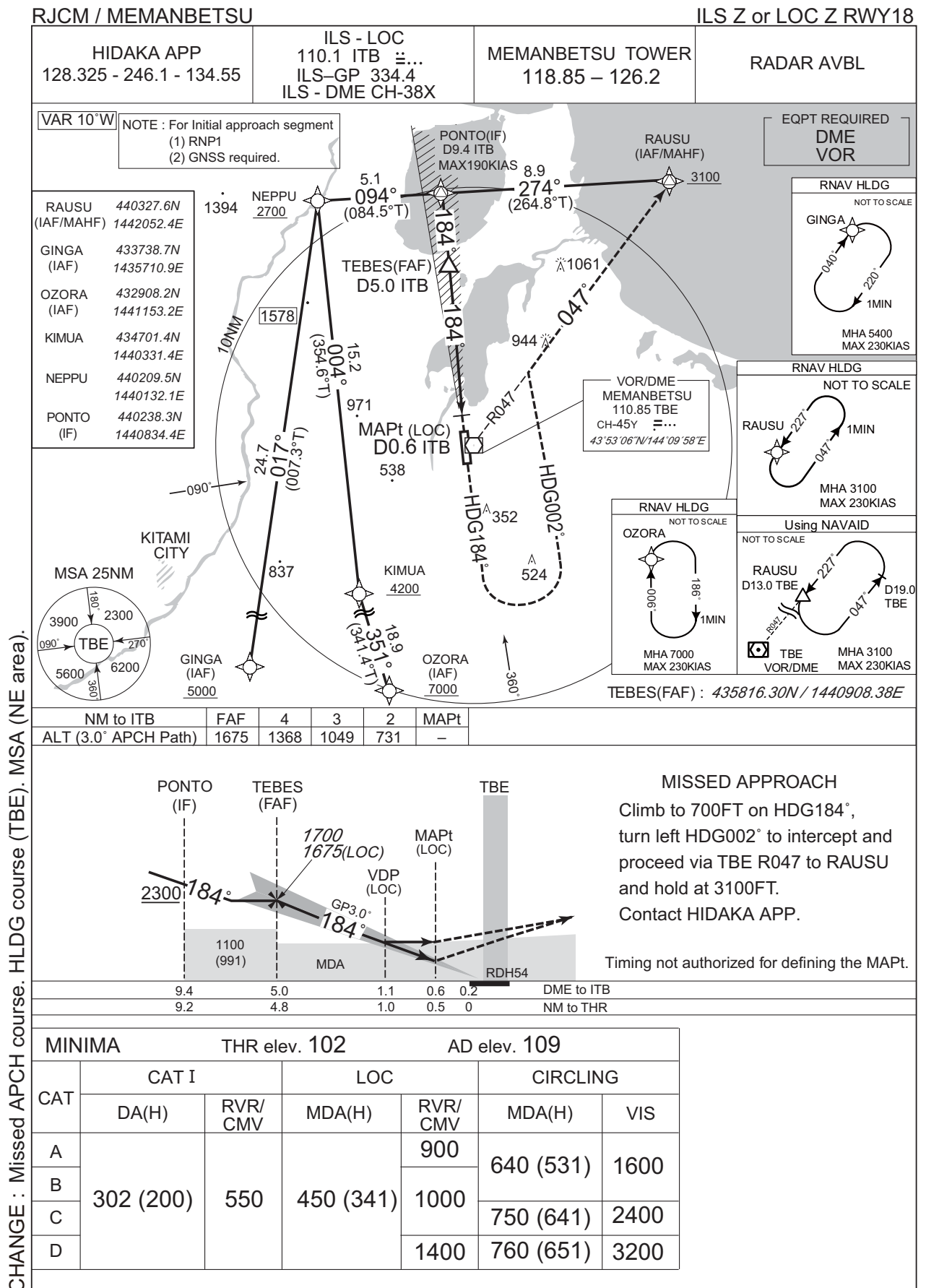
RNAV SID

Waypoint Coordinates

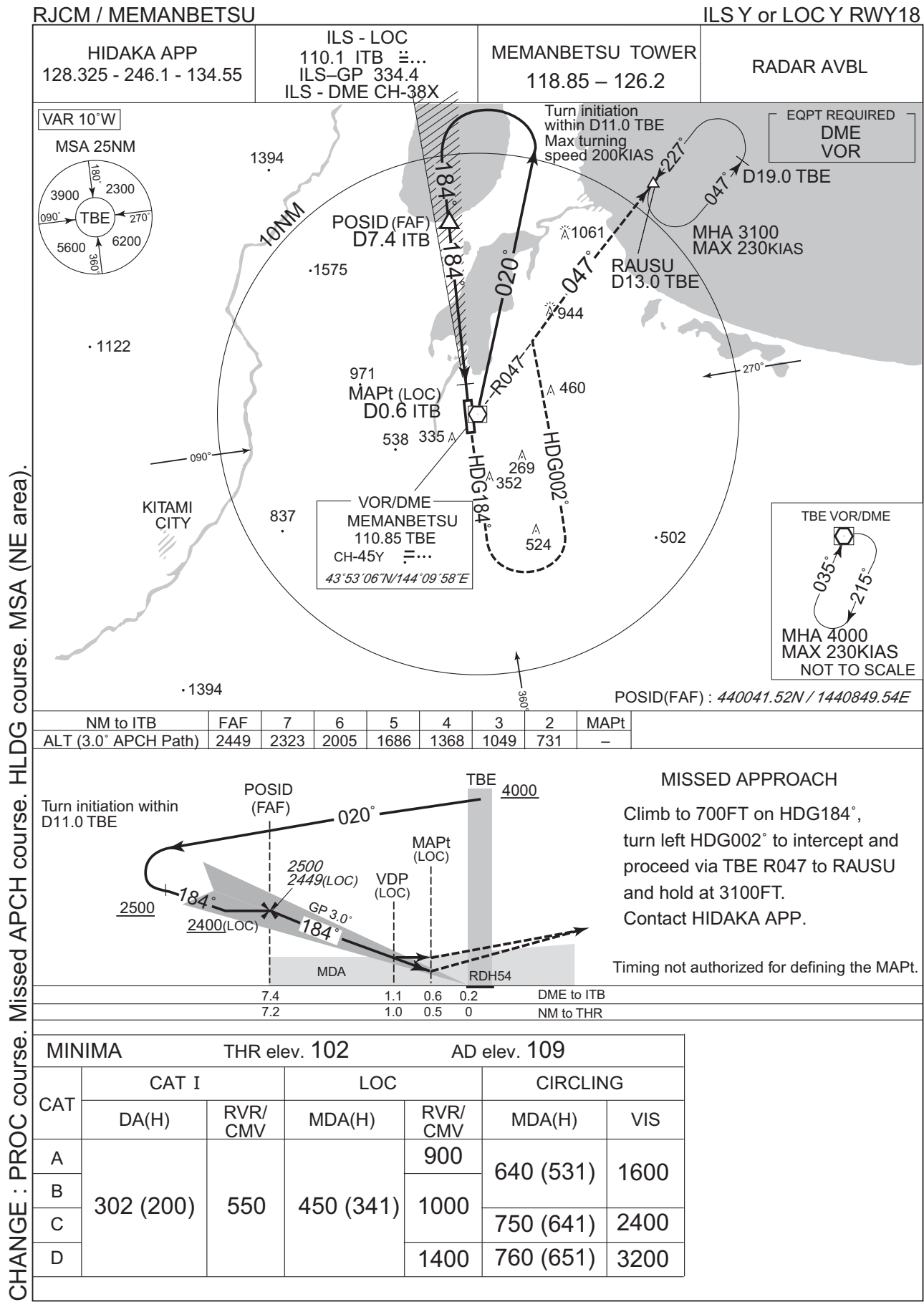
Waypoint Identifier	Coordinates
BISAK	434808.9N / 1441026.9E
AKNEP	434859.1N / 1435906.4E
NULKI	433301.6N / 1435313.7E

CHANGE : Waypoint Coordinates added.

INSTRUMENT APPROACH CHART

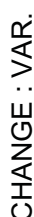
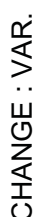


INSTRUMENT APPROACH CHART



RJCM / MEMANBETSU

ILS X RWY18



MINIMA		THR elev. 102	AD elev. 109	
CAT	CAT I		CIRCLING	
	DA(H)	RVR/ CMV	MDA(H)	VIS
A	302 (200)	550	640 (531)	1600
B				
C			750 (641)	2400
D			760 (651)	3200

INSTRUMENT APPROACH CHART

RJCM / MEMANBETSU

ILS X RWY18

Coding Table

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	GINGA	-	-	-9.6	-	-	+5000	-	-	RNP1
002	TF	KILOX	-	021 (011.9)	-9.6	17.2	-	-	-	-	RNP1
003	TF	CM853	-	009 (359.2)	-9.6	5.0	-	+2700	-200	-	RNP1
001	IF	OZORA	-	-	-9.6	-	-	+7000	-	-	RNP1
002	TF	KIMUA	-	351 (341.4)	-9.6	18.9	-	+4200	-	-	RNP1
003	TF	KILOX	-	002 (352.2)	-9.6	7.6	-	-	-	-	RNP1
004	TF	CM853	-	009 (359.2)	-9.6	5.0	-	+2700	-200	-	RNP1
001	IF	CM853	-	-	-9.6	-	-	+2700	-200	-	RNP1
002	RF Center: CMRF2 r=2.51NM	CM854	-	-	-9.6	3.8	R	+2200	-	-	RNP1
003	RF Center: CMRF2 r=2.51NM	CM855	-	-	-9.6	3.8	R	1700	-	-	RNP1
001	CA	-	-	184 (174.7)	-9.6	-	-	+700	-	-	RNP1
002	DF	RAUSU	-	-	-9.6	-	L	3100	-	-	RNP1
Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	Navigation Specification		
Hold	GINGA	040 (030.8)	-9.5	1.0 (-14000)	R	5400	FL140	-230 (-14000)	RNP1		
Hold	OZORA	006 (356.7)	-9.5	1.0 (-14000)	R	7000	FL140	-230 (-14000)	RNP1		
Hold	RAUSU	227 (217.0)	-9.5	1.0 (-14000)	L	3100	FL140	-230 (-14000)	RNP1		

Waypoint Coordinates

Waypoint Identifier	Coordinates	RF Arc Center Identifier	Coordinates
GINGA	433738.73N / 1435710.93E	CMRF2	435931.80N / 1440529.11E
OZORA	432908.24N / 1441153.19E		
KIMUA	434701.35N / 1440331.41E		
KILOX	435429.92N / 1440206.12E		
CM853	435929.83N / 1440200.57E		
CM854	440202.15N / 1440517.96E		
CM855	435945.90N / 1440856.76E		
RAUSU	440327.56N / 1442052.43E		

CHANGE : Navigation Specification(Basic RNP1 → RNP1).

RJCM / MEMANBETSU

HIDAKA APP
128.325 - 246.1 - 134.55

ILS - LOC
110.3 IHM
ILS - GP 335
ILS - DME CH-40X

MEMANBETSU TOWER
118.85 - 126.2

RADAR AVBL

VAR 10°W

MSA 25NM

TBE VOR/DME
110.85 TBE
CH-45Y
43°53'06"N/144°09'58"E

MAPt(LOC) D0.8 IHM
LOC OFFSET 1.7°

TSUKA(IAF) D18.0 TBE
5000

GINGA(IAF) D18.0 TBE
5000

MOHEA(IF) D15.3 IHM
R186/D16.0 TBE

ROMNY(FAF) D10.3 IHM
3278

OZORA(IAF) R186/D24.0 TBE
7000

Kitami City

10NM

EQPT REQUIRED
DME
VOR

NOT TO SCALE

	NM to IHM	MAPt	2	3	4	5	6	7	8	9	10	FAF
ALT (3.0° APCH Path)	-	756	1075	1393	1712	2030	2349	2667	2986	3304	3402	

MISSED APPROACH
Climb to 1000FT on HDG006°, turn left HDG225° to intercept and proceed via TBE R270 to TSUKA and hold at 5000FT. Contact HIDAKA APP.
Timing not authorized for defining the MAPt.

TBE

RDH54

MAPt(LOC)

VDP (LOC) (LOC) (LOC)

SDF (LOC) (LOC) (LOC)

ROMNY (FAF)

MOHEA (IF)

3500
3402(LOC)

2349
1075

910
(801)

1370
(1261)

3000
(2891)

4300

006°

	DME to IHM	NM to THR	0.2	0.8	1.4	3.0	7.0	10.3	15.3
			0	0.6	1.2	2.8	6.8	10.1	15.1

Missed APCH climb gradient MNM 3.0%

MINIMA
THR elev. 135
AD elev. 109

CAT	CAT I		LOC		CIRCLING	
	DA(H)	RVR/CMV	MDA(H)	RVR/CMV	MDA(H)	VIS
A	382 (247)	600	540 (431)	900	640 (531)	1600
B				1000		
C				750 (641)		
D				1400		

MINIMA with Missed APCH climb gradient of 2.5% are not established.

RJCM / MEMANBETSU

HIDAKA APP

128.325 - 246.1 - 134.55

MEMANBETSU VOR/DME

110.85 TBE
CH-45Y
43°53'06"N/144°09'58"E

MEMANBETSU TOWER

118.85 - 126.2

RADAR AVBL

VAR 10°W

MSA 25NM

1122

MHA 4000
MAX 230KIAS
NOT TO SCALE

NM to TBE

ALT (3.0° APCH Path)

FAF

2244

6

1926

5

1607

4

1289

3

970

2

652

MAPt

-

TBE 4000

018°

ABASI (FAF) 2244

181°

1400 (1291)

1100 (991)

600 (491)

MDA

MAPt

811

VDP

3.0° SDF

1607

181°

MISSED APPROACH

Climb to 900FT via TBE R181, turn left HDG002° to intercept and proceed via TBE R047 to RAUSU and hold at 3100FT. Contact HIDAKA APP.

Timing not authorized for defining the MAPt.

DME to TBE

NM to THR

MINIMA

THR elev. 102

AD elev. 109

CAT

MDA(H)

RVR/CMV

MDA(H)

VIS

A

480 (371)

900

640 (531)

1600

B

480 (371)

1000

750 (641)

2400

C

480 (371)

1000

750 (641)

2400

D

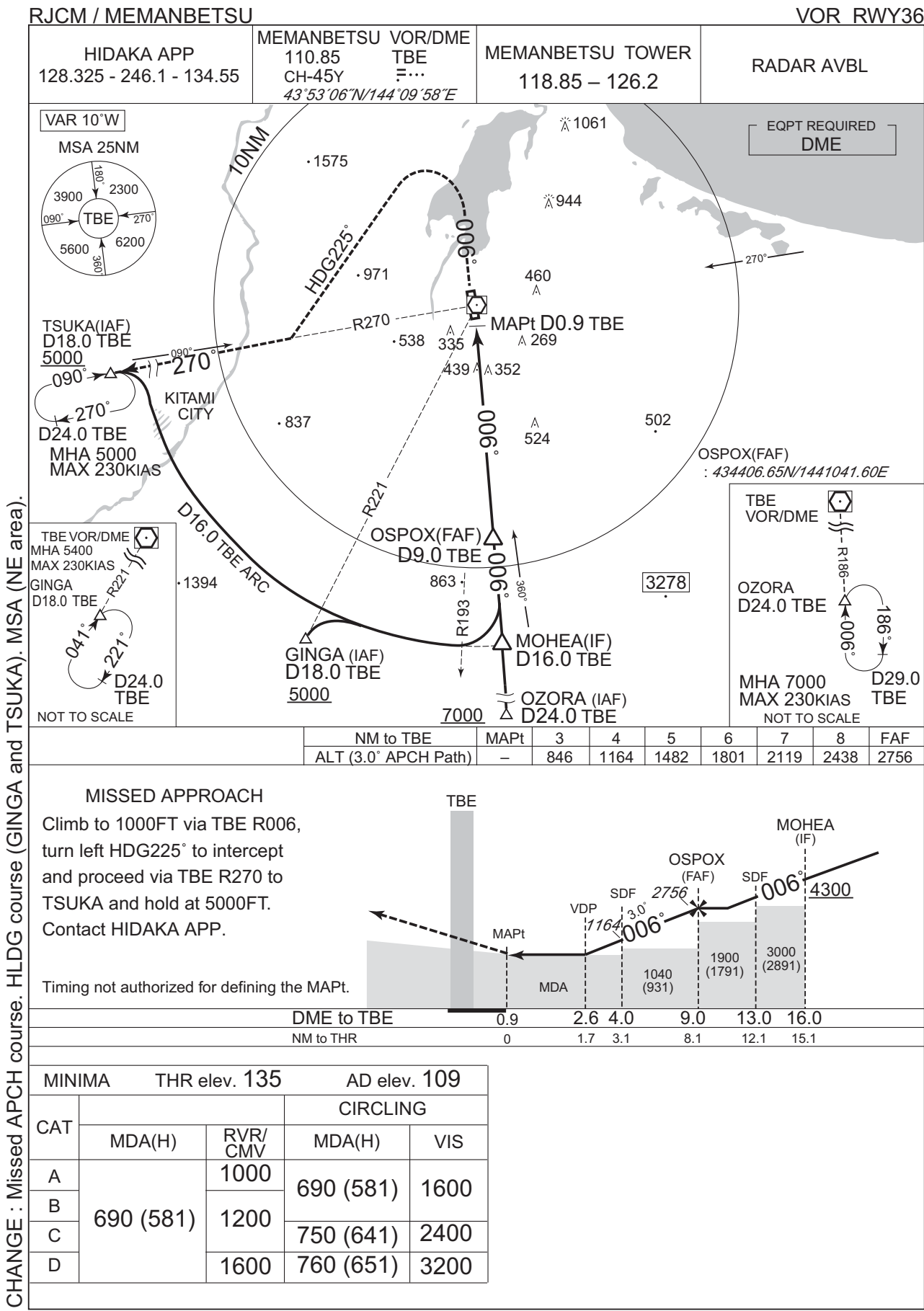
480 (371)

1400

760 (651)

3200

INSTRUMENT APPROACH CHART



RJCM / MEMANBETSU

HIDAKA APP

128.325 - 246.1 - 134.55

RNP APCH

MSAS CH83687
M18A

MEMANBETSU TOWER

118.85 – 126.2

RADAR AVBL

Baro-VNAV not authorized below -25°C

VAR 10°W

MSA 25NM

6200

ARP

ARP: 435250N/1440951E

NOT TO SCALE

GINGA

040°

220°

1MIN

MHA 5400

MAX 230KIAS

NOT TO SCALE

OZORA

006°

186°

1MIN

MHA 7000

MAX 230KIAS

NOT TO SCALE

RAUSU

221°

047°

1MIN

MHA 3100

MAX 230KIAS

1394

NEPPU

2700

1578

10NM

090°

24.7°

017°

(007.3°T)

15.2°

004°

(354.6°T)

18.9°

351°

(341.4°T)

184°

094°

(084.5°T)

5.1

184°

(174.7°T)

4.3

184°

(174.7°T)

4.9

274°

(264.8°T)

8.9

RAUSU

(IAF/MAHF)

3100

440327.56N

1442052.43E

432908.24N

1441153.19E

434701.35N

1440331.41E

440209.51N

1440132.13E

433738.73N

1435710.93E

440238.29N

1440834.36E

435820.17N

1440907.90E

435330.51N

1440945.38E

435031.32N

1441008.52E

NM to Next Fix

FAF

4

3

2

1

MAPt

ALT (3.0° APCH Path)

1700

1430

1111

793

475

-

PONTO

(IF)

2300

184°

1700

YOBIT

(FAF)

3.00°

184°

800

(691)

MDA

RDH54

0.9

0

NM to THR

Direct to CM852, turn left direct to RAUSU and hold at 3100FT. Contact HIDAKA APP.

MINIMA

THR elev.102

AD elev.109

CAT

LPV

LNAV/VNAV

LNAV

CIRCLING

DA(H)

RVR/CMV

DA(H)

RVR/CMV

MDA(H)

RVR/CMV

MDA(H)

VIS

A

352(250)

800

900

440(331)

900

640(531)

1600

B

360(258)

1000

1000

1000

750(641)

2400

C

369(267)

1200

1400

1400

760(651)

3200

D

379(277)

MINIMA with Missed APCH climb gradient of 2.5% are not established.

INSTRUMENT APPROACH CHART

RJCM / MEMANBETSU

RNP Z RWY18

FAS DATA BLOCK

Operation type	0	LTP/FTP ellipsoidal height	+00618
SBAS service provider identifier	2	FPAP latitude	435209.8350N
Airport identifier	RJCM	FPAP longitude	1440955.7670E
Runway	18	Threshold crossing height	00016.5
Approach performance designator	0	TCH units selector	1
Route indicator	Z	Glide path angle	03.00
Reference path data selector	0	Course width at threshold	105.00
Reference path ID	M18A	∠ length offset	0000
LTP/FTP latitude	435330.4920N	HAL	40.0
LTP/FTP longitude	1440945.3395E	VAL	50.0
CRC remainder	257C1F8B		

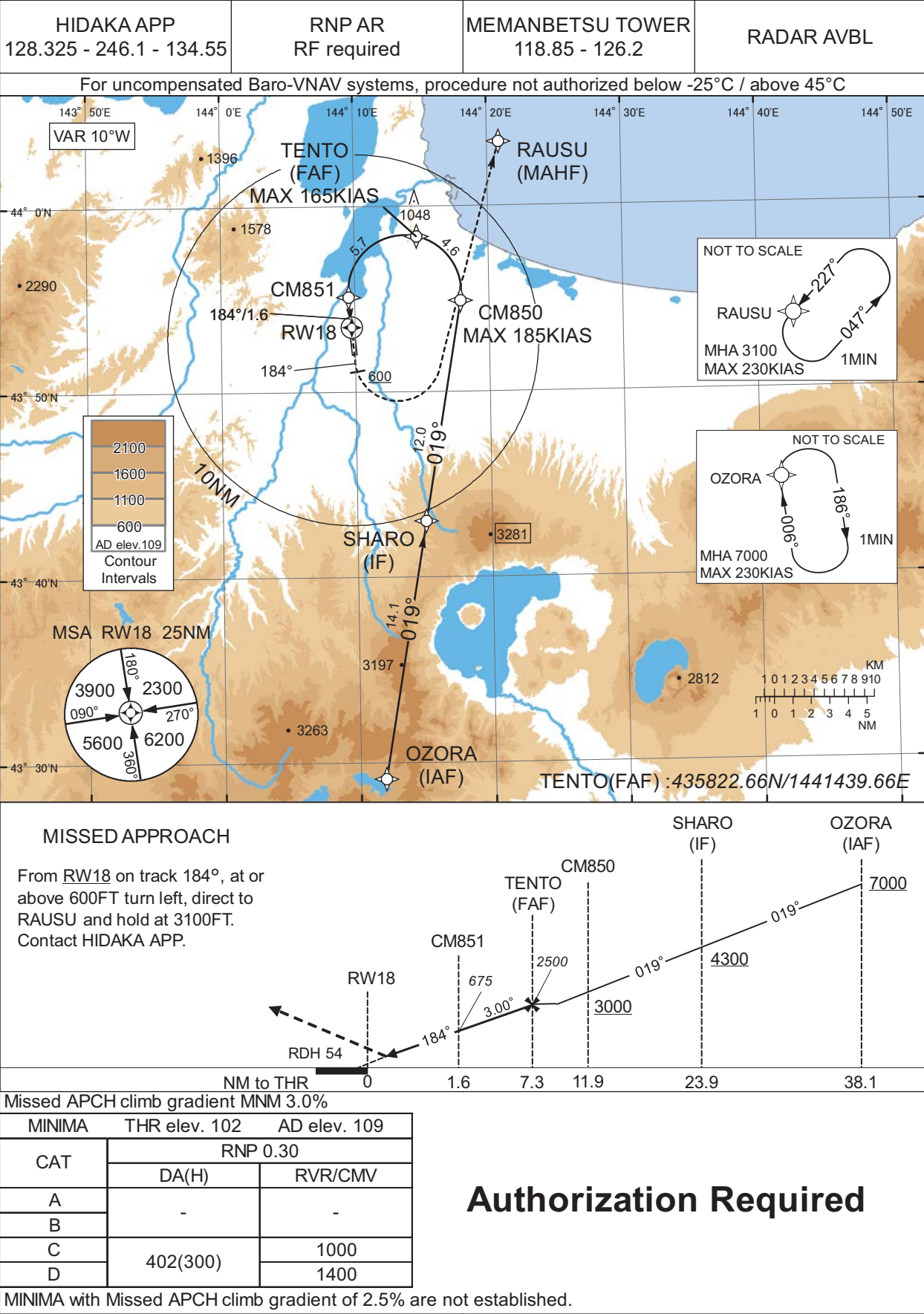
Required additional data

LTP/FTP orthometric height	31.0
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INSTRUMENT APPROACH CHART

RJCM / MEMANBETSU

RNP Y RWY18(AR)



INSTRUMENT APPROACH CHART

RJCM / MEMANBETSU

RNP Y RWY18(AR)

Coding Table

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	VPA/ RDH (°/FT)	RNP Value
001	IF	OZORA	-	-	-9.5	-	-	+7000	-	-	-
002	TF	SHARO	-	019 (009.5)	-9.5	14.1	-	+4300	-	-	1.0
003	TF	CM850	-	019 (009.6)	-9.5	12.0	-	+3000	-185	-	1.0
004	RF Center: CMRF1 r=3.04NM	TENTO	-	-	-9.5	4.6	L	2500	-165	-	1.0
005	RF Center: CMRF1 r=3.04NM	CM851	-	-	-9.5	5.7	L	675	-	-3.00	0.3
006	TF	RW18	Y	184 (174.7)	-9.5	1.6	-	156	-	-3.00/54	0.3
007	FA	-	-	184 (174.7)	-9.5	-	-	+600	-	-	1.0
008	DF	RAUSU	-	-	-9.5	-	L	3100	-	-	1.0

Path	Waypoint Identifier	Inbound Course °M(°T)	Magnetic Variation	Outbound Time (MIN)	Turn Direction	Minimum Altitude (FT)	Maximum Altitude (FT)	Speed (KIAS)	RNP Value
Hold	OZORA	006 (356.7)	-9.5	1.0 (-14000)	R	7000	FL140	-230 (-14000)	1.0
Hold	RAUSU	227 (217.0)	-9.5	1.0 (-14000)	L	3100	FL140	-230 (-14000)	1.0

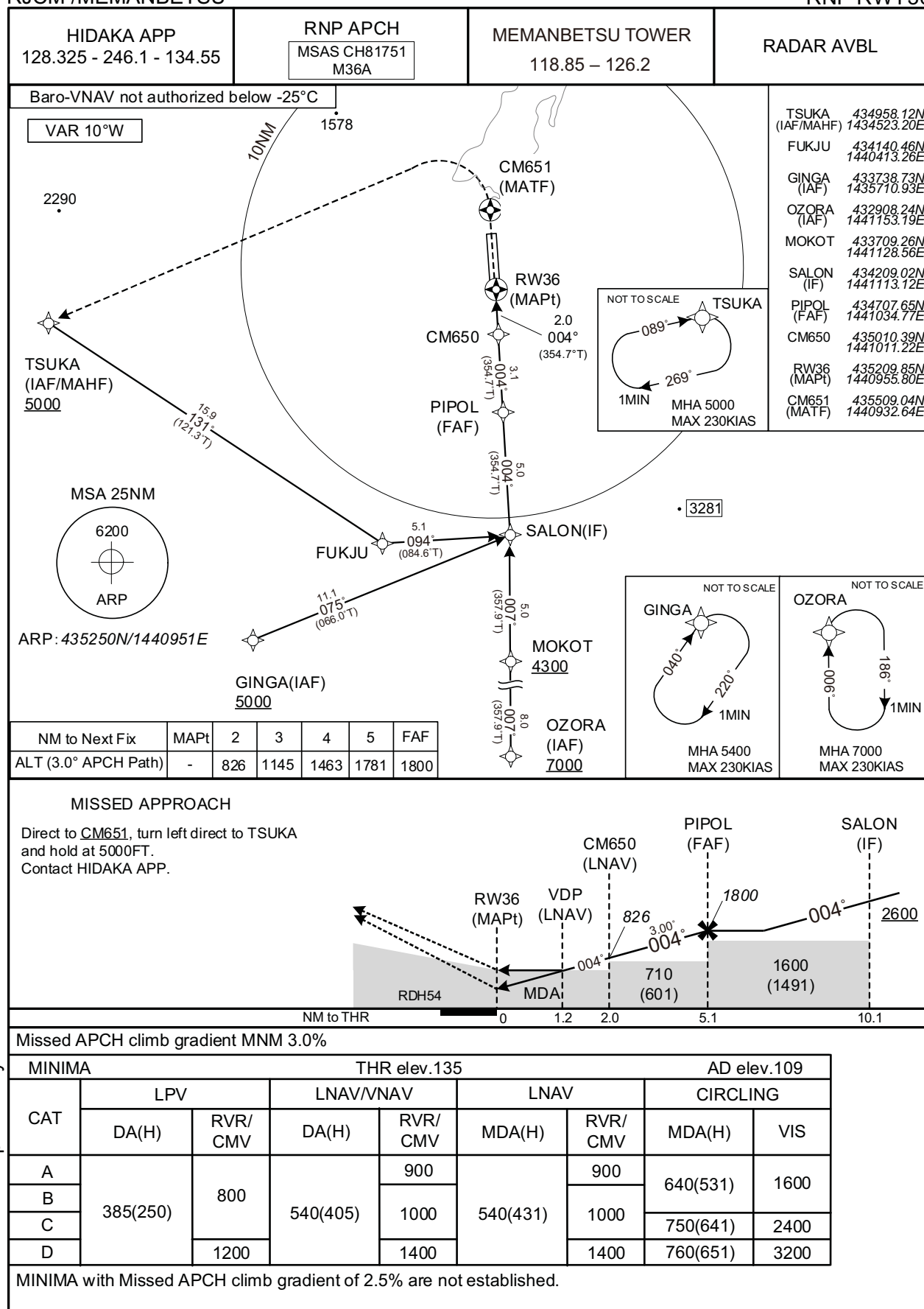
Waypoint Coordinates

Waypoint Identifier	Coordinates	RF Arc Center Identifier	Coordinates
OZORA	432908.24N / 1441153.19E	CMRF1	435524.66N / 1441344.15E
SHARO	434301.91N / 1441506.66E		
CM850	435454.27N / 1441753.05E		
TENTO	435822.66N / 1441439.66E		
CM851	435507.75N / 1440932.81E		
RW18	435330.51N / 1440945.38E		
RAUSU	440327.56N / 1442052.43E		

INSTRUMENT APPROACH CHART

RJCM /MEMANBETSU

RNP RWY36



CHANGE : Description of symbol for TSUKA.

INSTRUMENT APPROACH CHART

RJCM / MEMANBETSU

RNP RWY36

FAS DATA BLOCK

Operation type	0	LTP/FTP ellipsoidal height	+00719
SBAS service provider identifier	2	FPAP latitude	435330.4920N
Airport identifier	RJCM	FPAP longitude	1440945.3395E
Runway	36	Threshold crossing height	00016.5
Approach performance designator	0	TCH units selector	1
Route indicator		Glide path angle	03.00
Reference path data selector	0	Course width at threshold	105.00
Reference path ID	M36A	∠ length offset	0000
LTP/FTP latitude	435209.8350N	HAL	40.0
LTP/FTP longitude	1440955.7670E	VAL	50.0
CRC remainder	28C6AB24		

Required additional data

LTP/FTP orthometric height	41.1
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RJCM / MEMANBETSU

Visual REP



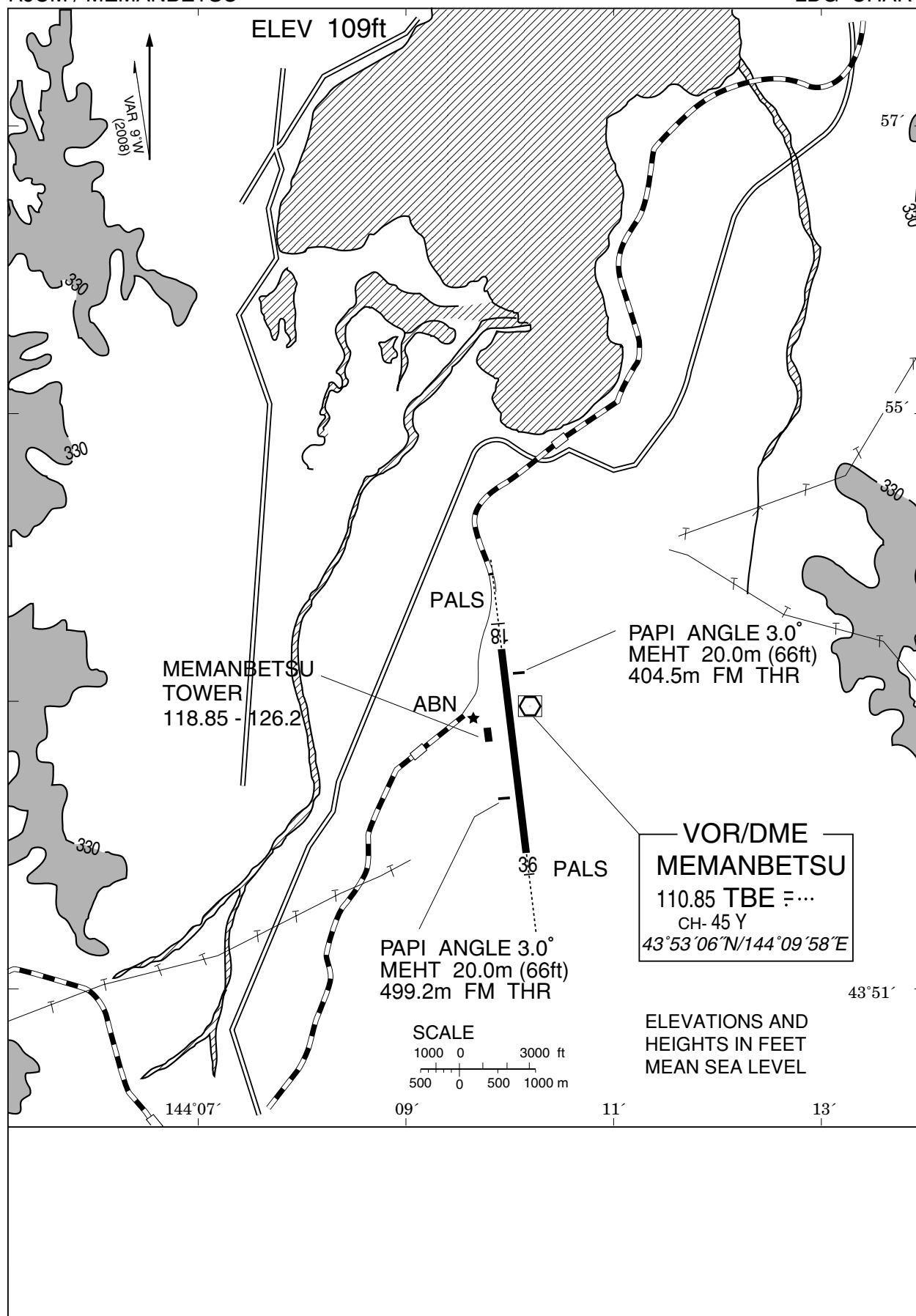
※図中に標高を示す数字がある場合、単位はメートル(m)である。The unit of measurement used to express elevation is meter(m).

CHANGE : Map updated. BRG/DIST from ARP.

Call sign	BRG / DIST from ARP	Remarks
網走 Abashiri	025°T / 9.2NM	JR駅 Station
湧湖 Tofutsuko	074°T / 10.5NM	湖 Lake
小清水 Koshimizu	097°T / 13.3NM	学校 School
東藻琴 Higashimokoto	110°T / 6.1NM	市街地 Town
美幌峠 Bihoro Toge	165°T / 14.4NM	峠 Pass
津別 Tsubetsu	210°T / 12.2NM	市街地 Town
美幌ステーション Bihoro Station	223°T / 3.7NM	JR駅 Station
北見 Kitami	248°T / 12.4NM	JR駅 Station
浜佐呂間 Hamasaroma	321°T / 15.5NM	佐呂間大橋 Bridge

RJCM / MEMANBETSU

LDG CHART



RJCM / MEMANBETSU

Minimum Vectoring Altitude CHART

